

Boolean logic

Igcse Computer Science

What is Boolean logic?

Boolean logic 布尔逻辑 works with values that are either **true** or **false**. In electronics these are shown as **1** (true) and **0** (false). A **logic gate** 逻辑门 takes one or more of these inputs and gives one output, following a fixed rule.

A **truth table** 真值表 lists every possible set of inputs and the output for each. You build it by writing out all the input combinations.

The six logic gates

You must know six gates. NOT has one input; all the others have two inputs (A and B).

NOT gate

The **NOT gate** 非门 reverses the input. Output is 1 when the input is 0.

A	Output
0	1
1	0

AND gate

The **AND gate** 与门 gives output 1 **only when both** inputs are 1.

A	B	Output
0	0	0
0	1	0
1	0	0
1	1	1

OR gate

The **OR gate** 或门 gives output 1 when **at least one** input is 1.

A	B	Output
0	0	0
0	1	1
1	0	1
1	1	1

NAND gate

The **NAND gate** 与非门 is AND followed by NOT. The output is the **opposite** of AND.

A	B	Output
0	0	1
0	1	1
1	0	1
1	1	0

NOR gate

The **NOR gate** 或非门 is OR followed by NOT. The output is the **opposite** of OR.

A	B	Output
0	0	1
0	1	0
1	0	0
1	1	0

XOR gate

The **XOR gate** 异或门 (exclusive OR) gives output 1 when the inputs are **different**.

A	B	Output
0	0	0
0	1	1
1	0	1
1	1	0

Logic expressions

A **logic expression** 逻辑表达式 writes a circuit using letters and gate words. The usual way to write the gates:

Gate	In words
NOT A	NOT A
A AND B	A AND B
A OR B	A OR B

For example, the expression (A AND B) OR (NOT C) means: do A AND B, do NOT C, then OR the two results together.

Logic circuits

A **logic circuit** 逻辑电路 joins gates together to carry out a task. The output of one gate can become the input of another. At IGCSE a circuit has up to **three inputs** and **one output**.

You must be able to move between three forms:

- a **problem statement** 问题陈述 (a description in words),
- a logic expression,
- a logic circuit,
- a truth table.

From a problem statement to a circuit

Read the statement and pick out the conditions and the logic words (and, or, not). For example:

An alarm (X) sounds when the door is open (A) AND the system is switched on (B).

This is $X = A \text{ AND } B$, so you draw one AND gate with inputs A and B.

Completing a truth table from a circuit or expression

To fill in a truth table:

1. Write **all** the input combinations. For three inputs there are 8 rows (000 up to 111).
2. Work out each gate's output in order, one column at a time.
3. The last column is the final output.

A	B	C	A AND B	(A AND B) OR C
0	0	0	0	0
0	0	1	0	1
0	1	0	0	0
0	1	1	0	1
1	0	0	0	0
1	0	1	0	1
1	1	0	1	1
1	1	1	1	1

Adding a middle "working" column for each gate makes the final output easy to fill in. Always draw the circuit exactly as the statement says, **without simplifying** it.